

When Can Training Data Be Recovered?

A rank condition for non-identifiability from the first-layer weight signal

A detailed, self-contained proof: if the gate matrix has rank below the number of training samples, the inputs cannot be recovered from the first-layer weight signal — no matter the algorithm.

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In plain terms — read this first

The reconstruction attack tries to read the training images back out of a trained network's weights. This note pins down a hard limit on when that is possible even in principle.

Every training sample leaves a fingerprint in the first layer's weights. But the fingerprint is filtered through how strongly each neuron reacts to that sample — a number we call the GATE. Collect all the gates into a matrix M : one row per neuron, one column per sample. The claim is:

The samples can be separated (recovered) only if M has rank at least N , the number of samples. If $\text{rank}(M) < N$, several samples share essentially the same gating pattern, their fingerprints add together into a blend, and no algorithm can un-mix them.

Intuition.

It is the same as being told $a + b = 5$ and asked for a and b . You cannot: (1,4), (2,3), (0,5) all fit — a whole line of answers, because there are two unknowns but only one equation. A low-rank gate matrix does exactly this: it gives you fewer independent measurements than samples, so several images collapse into one sum you can never split back apart. ' $\text{rank}(M)$ ' is just the number of genuinely independent measurements the neurons provide; when it drops below N , you are under-determined and recovery is impossible.

We (1) derive exactly what the attacker observes, (2) show it is a matrix product ' $\text{gates} \times \text{data}$ ', (3) prove the rank condition step by step, (4) count exactly how many indistinguishable alternative datasets exist, and (5) walk a tiny numeric example through every step by hand (§6).

Notation

d	input dimension (e.g. number of pixels).
N	number of training samples in the batch — the thing we count against.
m	number of hidden neurons (the width of the layer).
$\mathbf{x}_i \in \mathbb{R}^d$	the i -th training input; the columns of the data matrix X . This is what we recover.
$\mathbf{w}_k \in \mathbb{R}^d$	weight vector of neuron k ; the rows of the weight matrix W .
$\mathbf{v}_k \in \mathbb{R}$	output weight of neuron k (second layer).
σ, σ'	the activation function and its derivative.

$\langle \mathbf{w}_k, \mathbf{x}_i \rangle$	the pre-activation of neuron k on sample i (a scalar).
$\mathbf{M}_{ki}$	the GATE MATRIX entry: $\mathbf{M}_{ki} = \sigma'(\langle \mathbf{w}_k, \mathbf{x}_i \rangle)$.
$c_i \in \mathbb{R}$	a per-sample scalar coefficient coming from the loss (defined in §1).
Ω	the observed first-layer weight signal — an $m \times d$ matrix (the trained weights, or a fine-tuning weight change).

1. What the attacker sees, derived from scratch

We use the standard two-layer network of the reconstruction literature. With m hidden units and activation σ ,

$$f(\mathbf{x}; \theta) = \sum_{k=1}^m v_k \sigma(\langle \mathbf{w}_k, \mathbf{x} \rangle), \quad \mathbf{w}_k \in \mathbb{R}^d, \quad v_k \in \mathbb{R}.$$

Given the training set $\{(\mathbf{x}_i, y_i)\}$ and a per-sample loss ℓ , the empirical loss is $L(\theta) = \sum_i \ell(f(\mathbf{x}_i; \theta), y_i)$. The attacker observes the FIRST-LAYER weight signal. Where does that signal come from, and why is it a sum over the training data at all? Two scenarios produce the identical algebraic object, which is why one proof covers both.

Scenario A — the trained weights themselves (implicit bias / KKT).

Train a homogeneous net to convergence on a classification loss. Gradient descent's velocity is a NON-NEGATIVE combination of the per-sample gradients (each coefficient $\alpha_i = e^{-y_i f(\mathbf{x}_i)} > 0$):

$$-\nabla_{\theta} L = \sum_i \alpha_i y_i \nabla_{\theta} f(\mathbf{x}_i), \quad \alpha_i = e^{-y_i f(\mathbf{x}_i)} > 0,$$

so the weights can never leave the cone those N gradients span. In the limit they converge to a KKT point of the max-margin problem — the final parameters are literally a weighted sum over the training points:

$$\theta = \sum_i \lambda_i y_i \nabla_{\theta} f(\mathbf{x}_i; \theta), \quad \lambda_i \geq 0 \quad (\text{nonzero only on support vectors}).$$

The attacker HAS θ (public model), so θ is a known sum over the data; here the per-sample scalar is $c_i = \lambda_i y_i$.

Scenario B — a fine-tuning update (gradient inversion / LoRA).

The attacker sees a weight change $\Delta \mathbf{W} = -\eta \nabla L$ from one fine-tuning step (or a LoRA adapter encoding it), directly proportional to the gradient; here $c_i = \ell'(f(\mathbf{x}_i), y_i)$, the loss slope.

Both scenarios give $\Omega = \sum_i c_i \cdot$ (first-layer gradient of f at $\mathbf{x}_i$), with c_i a nonzero per-sample scalar. The proof below uses ONLY that c_i is a nonzero scalar per sample, so it covers both attacks at once. We now derive that per-sample gradient.

The ingredient both scenarios share is the per-sample gradient of the network OUTPUT, $\partial f(\mathbf{x}_i) / \partial \mathbf{w}_k$ (NOT the loss gradient — in Scenario A the loss gradient is ≈ 0 at convergence; the signal is the weights themselves). By the chain rule — only the k -th term of f depends on $\mathbf{w}_k$, and $\partial \langle \mathbf{w}_k, \mathbf{x}_i \rangle / \partial \mathbf{w}_k = \mathbf{x}_i$:

$$\frac{\partial f(x_i)}{\partial w_k} = v_k \sigma'(\langle w_k, x_i \rangle) x_i.$$

So in EITHER scenario, row k of the observed signal Ω is a c_i -weighted sum of these per-sample gradients, $\Omega_{k,:} = \sum_i c_i \partial f(x_i) / \partial w_k$, with (k, l) entry (row k = neuron, column l = input coordinate)

$$\Omega_{k,l} = \sum_{i=1}^N c_i v_k \sigma'(\langle w_k, x_i \rangle) x_{i,l}.$$

(Note Ω is the WEIGHT SIGNAL — the trained weights in Scenario A, the weight change in Scenario B — not the loss gradient $\partial L / \partial W$, which is only Scenario B's special case, $c_i = -\eta \ell'$.) The one fact used below: c_i is a single scalar per sample, the SAME for every neuron k (it carries no k index).

Intuition.

Each neuron reports a weighted sum of the training images. The weight it puts on image x_i is its gate $\sigma'(\langle w_k, x_i \rangle)$ — how sensitive that neuron is to that image. The image only ever enters the observation through these gated sums. So the question 'can we recover the images?' becomes 'can we undo these gated sums?' — a linear-algebra question.

2. The observation is 'gates $\times$ data'

Collect the pieces into matrices. Let

$$X = [x_1, \dots, x_N] \in \mathbb{R}^{d \times N} \text{ (columns are the samples),} \quad M_{ki} = \sigma'(\langle w_k, x_i \rangle) \in \mathbb{R}^{m \times N},$$

$$D_v = \text{diag}(v_1, \dots, v_m), \quad D_c = \text{diag}(c_1, \dots, c_N), \quad G := D_v M D_c \in \mathbb{R}^{m \times N}.$$

Why exactly these three pieces: in each term $c_i \cdot v_k \cdot \sigma'(\langle w_k, x_i \rangle) \cdot x_i$, the factor v_k depends on the NEURON only, c_i on the SAMPLE only, and the gate $\sigma'(\langle w_k, x_i \rangle)$ is the ONLY factor coupling a neuron k to a sample i . The two separable factors become the diagonals — v_k scales row k (the left diagonal), c_i scales column i (the right diagonal) — and the coupled gate is M . Equivalently, Ω is a sum of rank-1 outer products (g_i = column i of G):

$$\Omega = \sum_{i=1}^N g_i x_i^\top = G X^\top, \quad (g_i)_k = v_k c_i \sigma'(\langle w_k, x_i \rangle),$$

which is the outer-product form of the matrix product: stack the g_i as columns of G and the x_i as columns of X . This is proved index-by-index next.

Lemma 1 (the observation factorizes).

The whole observation is a single matrix product of the gates and the data:

$$\Omega = G X^\top.$$

Proof. Compare entry (k, l) of both sides. On the right,

$$(GX^\top)_{k,l} = \sum_{i=1}^N G_{ki} (X^\top)_{i,l} = \sum_{i=1}^N v_k M_{ki} c_i x_{i,l} = v_k \sum_{i=1}^N c_i \sigma'(\langle w_k, x_i \rangle) x_{i,l},$$

which is exactly the entry $\Omega_{k,l}$ from §1. Since this holds for every k and l , $\Omega = G X^\top$. ■

Dimension check.

Every term has a clear shape, and they compose consistently (m = neurons, N = samples, d = input dimension):

- each sample $x_i \in \mathbb{R}^d$ is a d -vector; the data matrix X is $d \times N$; so $X^\top$ is $N \times d$.
- each $w_k \in \mathbb{R}^d$, and the first-layer weight matrix W is $m \times d$.
- the gate $\sigma'(\langle w_k, x_i \rangle)$ is a scalar, so the gate matrix M is $m \times N$.
- the two diagonal scalings are $m \times m$ and $N \times N$, so G stays $m \times N$.
- $\Omega = G X^\top$ multiplies an $m \times N$ by an $N \times d$ — the inner N 's cancel — giving an $m \times d$ matrix, the same shape as the weight matrix W (as it must be — Ω lives in weight space).

$$\Omega (m \times d) = G (m \times N) \cdot X^\top (N \times d)$$

Lemma 2 (rescaling does not change rank).

Assume (H3): every $v_k \neq 0$ and every $c_i \neq 0$. Then the two diagonal scalings are invertible, and multiplying by an invertible matrix never changes rank, so

$$\text{rank}(G) = \text{rank}(D_v M D_c) = \text{rank}(M).$$

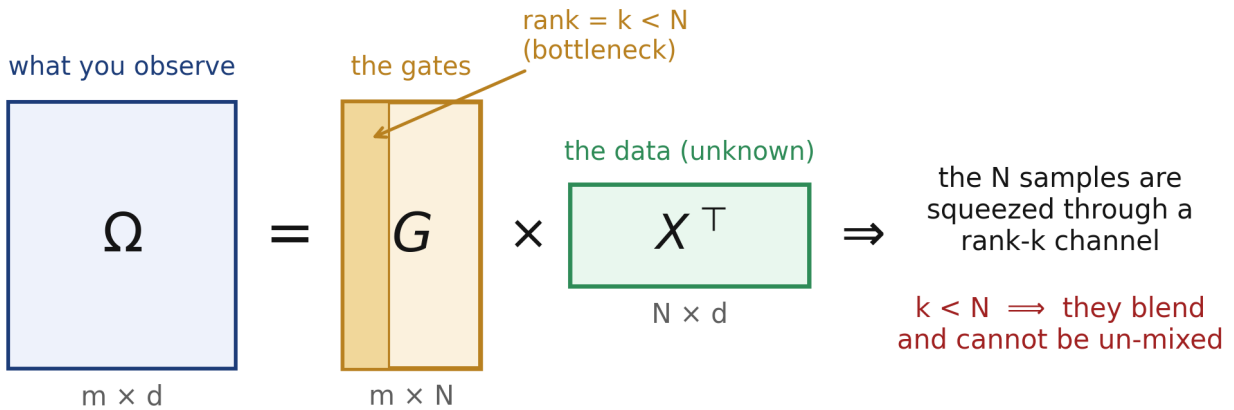


Figure 1. The observation Ω is the product of the gate matrix G ($m \times N$, but only rank k) and the data $X^\top$ ($N \times d$). The N samples must pass through a channel of rank k . When $k < N$ the channel is too narrow to carry them separately — they arrive blended.

3. Three linear-algebra facts we will use

The proof needs only these standard facts; we state them plainly so the argument is self-contained.

(i) Solution sets of linear equations. If a linear map Φ satisfies $\Phi(X) = \Omega$ and also $\Phi(X') = \Omega$, then $\Phi(X' - X) = 0$. So every solution X' equals the true X plus something the map sends to zero. In symbols, the full solution set is $X + \ker \Phi$, where $\ker \Phi = \{H : \Phi(H) = 0\}$ is the KERNEL (null space).

(ii) Rank-nullity. For a linear map given by a matrix A with n columns, $\dim(\ker A) = n -$

$\text{rank}(A)$. The rank counts the independent directions the map keeps; the nullity $n - \text{rank}(A)$ counts the directions it destroys.

(iii) Positive dimension means infinitely many. A subspace of dimension ≥ 1 contains a whole line's worth of points — uncountably many. So if the solution set $X + \ker \Phi$ has dimension ≥ 1 , there are infinitely many datasets consistent with the same observation.

Intuition.

Whatever the map 'destroys' (its kernel) is information you can never recover from the output. If the data can be changed along any kernel direction without changing Ω , then Ω simply does not determine the data. The size of the kernel is the size of your ignorance.

4. What 'recoverable' means, precisely

Fix the gate values M and the scalars v, c — i.e. treat the gates as GIVEN (assumption H2; §8 explains why this is the right and strongest way to phrase the limit). Under H2 the map from data to observation is linear:

$$\Phi : \mathbb{R}^{d \times N} \rightarrow \mathbb{R}^{m \times d}, \quad \Phi(X) = G X^\top.$$

We say the data X is IDENTIFIABLE from Ω if the only datasets X' with $\Phi(X') = \Omega$ are X itself and its column permutations. Column permutation is a benign symmetry — it only relabels which image we call 'the first' — so it does not count as a failure. NON-identifiability means some genuinely different dataset (not a relabeling) produces the identical observation; then no procedure, however powerful, can tell which one was the true training set.

5. The theorem and its proof

Theorem (rank obstruction).

Assume (H1) $\Omega = G X^\top$, (H2) the gates G are fixed, (H3) $v_k, c_i \neq 0$. If $\text{rank}(M) = k < N$, then the set of datasets consistent with the observation is an affine subspace

$$\Phi^{-1}(\Omega) = X + K, \quad K = \{H \in \mathbb{R}^{d \times N} : \text{every row of } H \text{ lies in } \ker G\},$$

$$\dim K = d(N - k) \geq d \geq 1.$$

In words: there is a whole $d(N-k)$ -dimensional family of different datasets that all produce the same Ω . Hence X is not identifiable.

Proof, in five steps.

Step 1 (the solution set is $X + \text{kernel}$). Φ is linear (Lemma 1: $\Phi(X) = G X^\top$). By fact (i), the datasets consistent with Ω are exactly $\Phi^{-1}(\Omega) = X + \ker \Phi$, since the true X is one solution.

Step 2 (what is in the kernel?). We find every H with $\Phi(H) = G H^\top = 0$. Fix a column index l and look at column l of $G H^\top$:

$$(G H^\top)_{k,l} = \sum_{i=1}^N G_{ki} H_{l,i} = (G \cdot (\text{row } l \text{ of } H)^\top)_k.$$

This is zero for all neuron indices k exactly when G times (row l of H)^T is the zero vector — that is, when row l of H lies in $\ker G$. This must hold for each of the d rows of H , and the rows are unconstrained otherwise, so

$$\ker \Phi = \{H : \text{each of the } d \text{ rows of } H \in \ker G\}.$$

Step 3 (count the dimensions). Each of the d rows ranges independently over the space $\ker G$, so the dimension of the kernel is d copies of $\dim(\ker G)$. By rank-nullity (fact ii) applied to G , which has N columns,

$$\dim \ker \Phi = d \cdot \dim(\ker G) = d(N - \text{rank } G) = d(N - k),$$

using $\text{rank } G = \text{rank } M = k$ from Lemma 2.

Step 4 (the family is nontrivial). Because $k < N$ we have $N - k \geq 1$, so $\dim \ker \Phi = d(N - k) \geq d \geq 1$. By fact (iii) this is a positive-dimensional space: it contains infinitely many distinct H , hence $\Phi^{-1}(\Omega) = X + \ker \Phi$ contains infinitely many distinct datasets, all producing the identical Ω .

Step 5 (they are genuinely different, not relabelings). The benign symmetries are the $N!$ column permutations (plus any finite sign/scale symmetries) — a FINITE set. A finite set cannot fill a positive-dimensional continuum, so all but a measure-zero portion of the consistent datasets are unrelated to X by any symmetry. Therefore no function of Ω can single out the true X . ■

6. A worked example, walked through step by step

Smallest interesting world: 2 training images, each with $d = 2$ pixels, and $m = 2$ neurons.

Step 0 — the setup.

The two secret images the attacker wants are $x_1 = (1, 0)$ and $x_2 = (0, 1)$. The gate matrix G has entry $G[k, i] = \text{how strongly neuron } k \text{ reacts to image } i \text{ (times the loss weight)}$; take

$$G = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \Rightarrow \text{row } 2 = 2 \times \text{row } 1 : \text{rank}(G) = 1 < 2 = N.$$

Notice the two neurons react to the images in the SAME proportion (1 : 2) — they are not two independent viewpoints, really just one. That is what $\text{rank}(G) = 1$ means, and it is the whole problem.

Step 1 — what the attacker sees.

The observed weights are $\Omega = G X^T$, which is the same as one fingerprint per image — each image imprinted by how the neurons react to it ($g_i = \text{column } i \text{ of } G$, so $g_1 = (1, 2)$, $g_2 = (2, 4)$):

$$\Omega = g_1 x_1^\top + g_2 x_2^\top = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Step 2 — the collapse (why it breaks).

Because $g_2 = 2 \cdot g_1$, factor g_1 out of the sum:

$$\Omega = g_1 x_1^\top + (2g_1) x_2^\top = g_1 (x_1 + 2x_2)^\top.$$

The two images have merged: Ω depends on them ONLY through the single combination $s := x_1 + 2x_2$. Reading it off Ω gives $s = (1, 2)$ — and that is ALL the attacker can ever learn. One equation, two unknown images. (This is the $a + b = 5$ situation, exactly.)

Step 3 — infinitely many datasets fit.

Any (x_1', x_2') with $x_1' + 2x_2' = (1, 2)$ gives the identical Ω . Choose x_2' freely; x_1' is then forced. For example $x_2' = (-1, 0)$ forces $x_1' = (1, 2) - 2(-1, 0) = (3, 2)$. Check that this completely different dataset reproduces the weights:

$$GX'^\top = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \Omega$$

$\{(1,0), (0,1)\}$ and $\{(3,2), (-1,0)\}$ — two datasets with nothing in common — produce the IDENTICAL Ω . The attacker cannot tell which was the training set. That is non-recoverability, concretely.

Step 4 — this family IS the kernel.

The difference between the impostor and the truth is $(x_1' - x_1, x_2' - x_2) = ((2,2), (-1,-1))$. Its effect on the only thing Ω sees: $\Delta s = (2,2) + 2(-1,-1) = (0, 0)$ — invisible. The set of all such invisible differences (h_1, h_2) with $h_1 + 2h_2 = 0$ is the KERNEL: h_2 is free in $\mathbb{R}^2$ (2 numbers), $h_1 = -2h_2$ is forced, so its dimension is 2 — matching the formula $d(N - \text{rank}) = 2(2 - 1) = 2$. The 'maneuver' is just: count the directions you can move the data without changing the measurement.

Step 5 — the contrast that shows rank is everything.

Had the neurons given INDEPENDENT views — say

$$G = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (\text{rank } 2 = N)$$

— then $\Omega = g_1 x_1^\top + g_2 x_2^\top$ with $g_1 = (1,0)$, $g_2 = (0,1)$ hands you the images directly: row 1 of Ω is x_1 and row 2 is x_2 . Unique recovery. Nothing collapsed because g_2 is not a multiple of g_1 . $\text{rank} = N \Rightarrow \text{recoverable}$; $\text{rank} < N \Rightarrow \text{not}$. That is the theorem.

7. Why low rank blurs the samples (the mechanism)

When two samples have proportional gate columns (the rank-1 case), every neuron reacts to them in lockstep — whatever neuron k does to x_1 , it does a fixed multiple of to x_2 . The observation then depends on the two images only through one fixed linear combination of them, say $s = c_1x_1 + \gamma c_2x_2$. The individual images are free to slide along the line that keeps s fixed, and every point on that line fits the data equally well. The reconstruction can at best return the blend s — never the two images apart. This is exactly the 'superposition' failure seen empirically for $N \geq 2$ once the per-sample gradients become collinear, and it is why raising the gate rank (through the activation, the anchor, or the LoRA rank) is what buys separability.

8. Corollary, scope, and honest caveats

Corollary (a necessary width).

Recovering X from the first-layer signal requires $\text{rank}(M) \geq N$. Since $\text{rank}(M) \leq \min(m, N)$, it in particular requires the hidden width $m \geq N$: a layer narrower than the batch cannot expose its samples through this signal, whatever the activation, optimizer, or algorithm.

(a) Why 'fixed gates' is the right framing. In reality the gates depend on the data ($M_{ki} = \sigma'(\langle w_k, x_i \rangle)$), so the true map $X \mapsto \Omega(X) = G(X) X^T$ is bilinear, and the alternative datasets X' generally have different gates. The theorem therefore describes the STRONGEST possible attacker — one additionally handed the exact gate values of the true data. A real attacker knows less, so cannot do better on this count. What is rigorously proved: the first-order (gate-factorization / NTK-linearized) information in Ω is INSUFFICIENT when $\text{rank}(M) < N$, and that first-order regime is exactly where the attack operates. The extra self-consistency the true gates must satisfy can in special cases shrink the family, so this is a necessary condition, not a claim that no nonlinear trick could ever help.

(b) Necessary vs. sufficient. $\text{rank}(M) \geq N$ is necessary. In the fixed-gate model it is also generically sufficient: if $\text{rank}(G) = N$ then $X = (G^+ \Omega)^T$ is the unique solution ($G^+ =$ Moore–Penrose pseudoinverse). Turning this into a clean sufficiency theorem for the full bilinear problem — exactly when ΔW determines $\{x_i\}$ uniquely — is the open identifiability question the thesis targets.

(c) One layer. We used the first-layer factorization, where the data appears explicitly (the per-sample gradient $\partial f / \partial w_k$ carries x_i). Deeper layers may add constraints; the bound is a lower bound on difficulty, tight in the linearized regime.

9. Why this matters for the thesis

$\text{rank}(M)$ is a hard ceiling on how many samples can be separated from the first-layer signal, and the three levers of the study act on it directly. The ACTIVATION determines σ' and hence the entries of M . The ANCHOR — linearizing partway between the initial weights θ_0 and the fine-tuned weights (the α -sweep below) — shifts the pre-activations $\langle w_k, x_i \rangle$ and so reshapes M . A LoRA adapter replaces Ω by a rank- r projection of it, tightening the ceiling further to

$$\text{leakage} \leq \min(\text{rank}(M), r, N), \quad \theta_{\text{anchor}} = (1 - \alpha) \theta_0 + \alpha \theta_T.$$

The theorem above is the first, information-theoretic, of these ceilings — the reason 'how many samples, and through what activation and rank' is the right axis for the whole study.